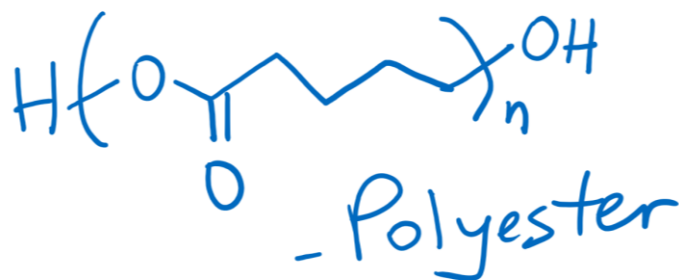


Question Set 4

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

- 1) Monomer I produces a polymer when treated with H_2SO_4 at high temperatures.
- a) Draw the structure of the polymer formed and indicate what type of polymer it is.



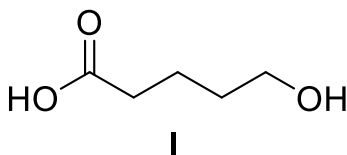
- b) Calculate the maximum degree of polymerisation (DP) that you can achieve if the conversion is 90 %. Show your calculations clearly and indicate what equation you use to calculate it.

ETQ: A full answer for full marks in an exam will explain the following: This is a step-growth polymerisation, therefore, you can use the Carothers equations to determine degree of polymerisation (DP). Applying Carothers equation (show equation and explain the variables):

$$\overline{DP} = \frac{1}{1-p}$$

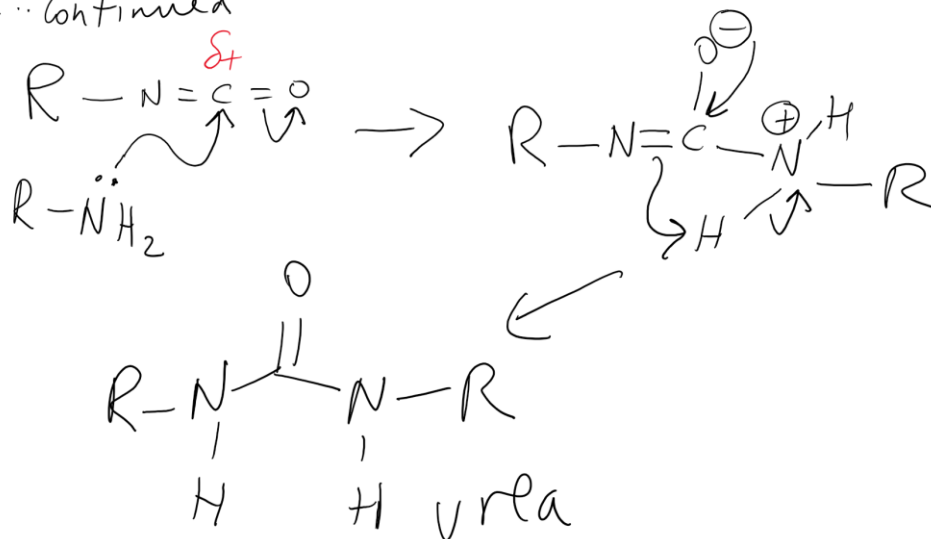
Conversion 90%: $p=0.9$, therefore (show calculation): $DP=10$.

- c) Draw the mechanism of the reaction

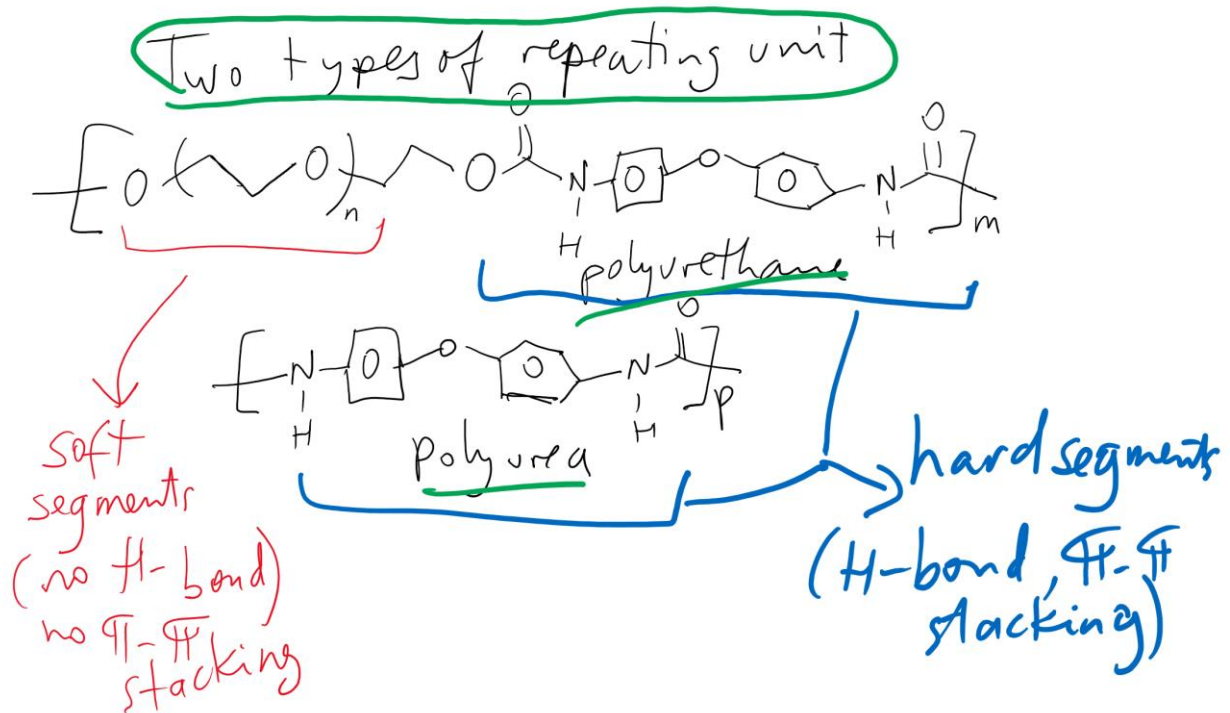


Please see the mechanism video to understand the mechanism. This would be a mechanism video from Molecules to Materials (Fischer esterification)

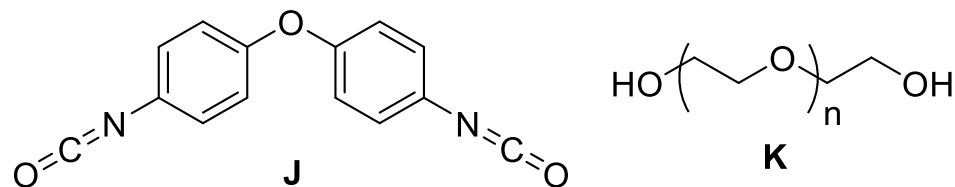
... continued



- b) Draw a structure of the final polymer that clearly shows all the types of repeating units and functional groups present in the final material.



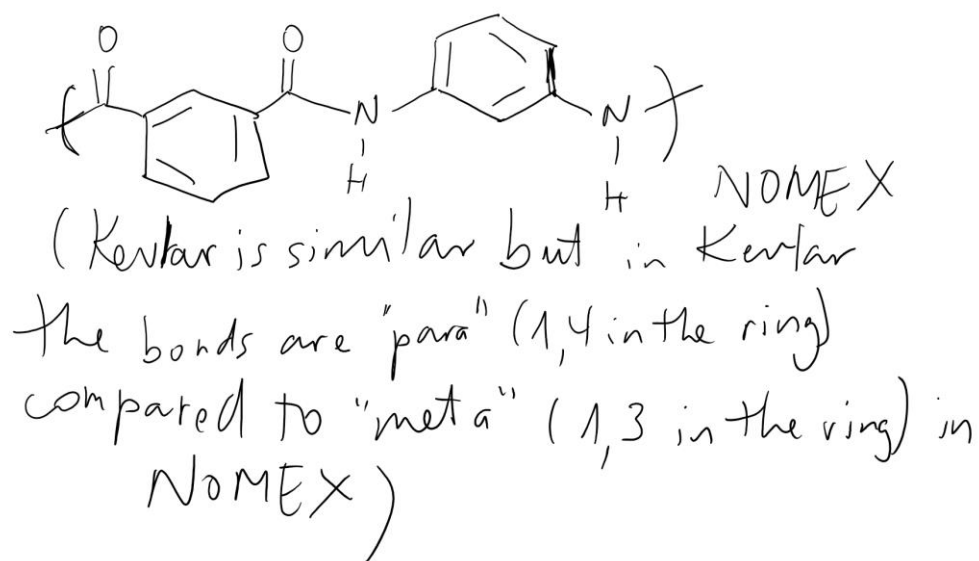
- c) What type of properties would you expect from a polymer such as described in (b)? Explain with reference to the structure of the polymer.



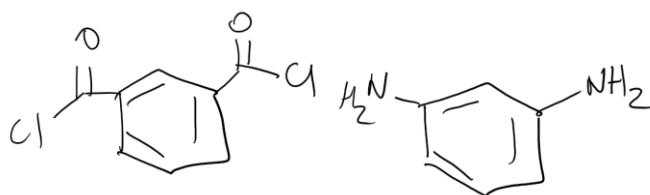
c) The fact that the diol, K, is polymeric, acts by creating segments where there is no H-bond. Because the material has both soft and hard segments (see annotations on the structure), the soft segments are more elastomeric ("stretchy") in nature, while the highly associated hard segments (attracted to each other by H-bonding mostly, with some elements of pi-pi stacking), which act as "physical crosslinks" restricting the stretching. This is seen in materials like Spandex or Lycra, stretchable fabrics.

3) Nomex® is an aramid, like Kevlar®.

a) What is the structure of Nomex®? (You will have to find this online or in a textbook)

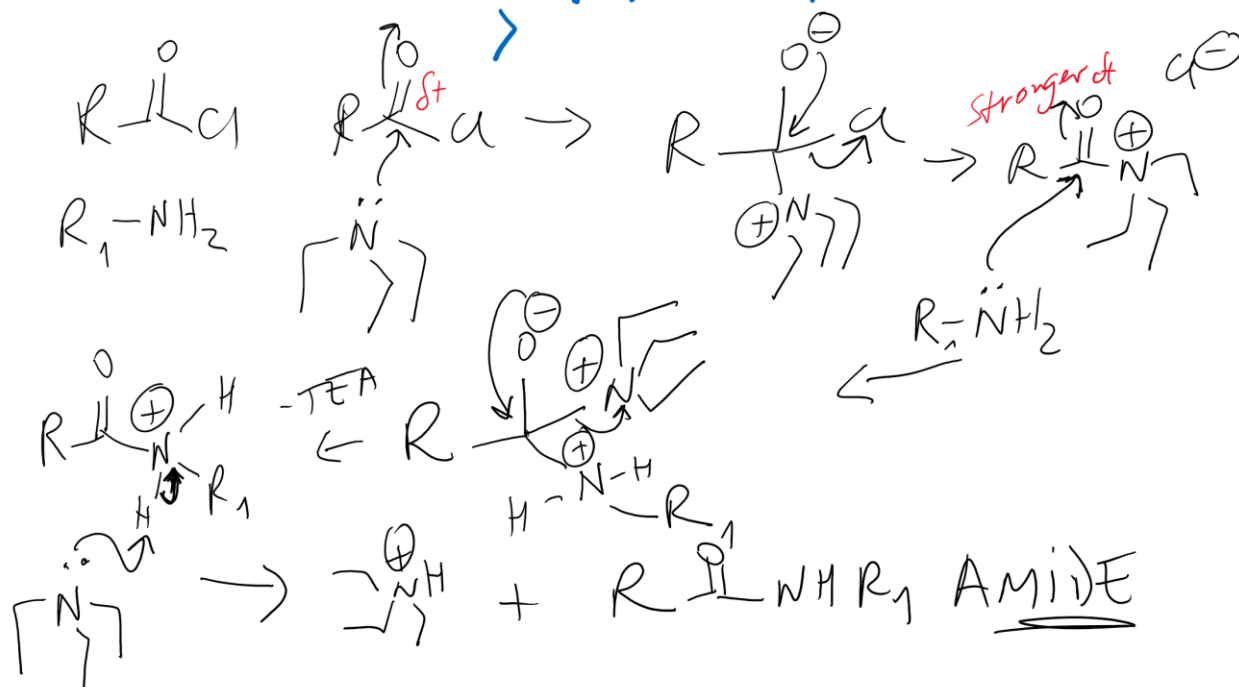


b) Propose the structure of a suitable bis-acid chloride and a suitable bis-amine to synthesise Nomex®.



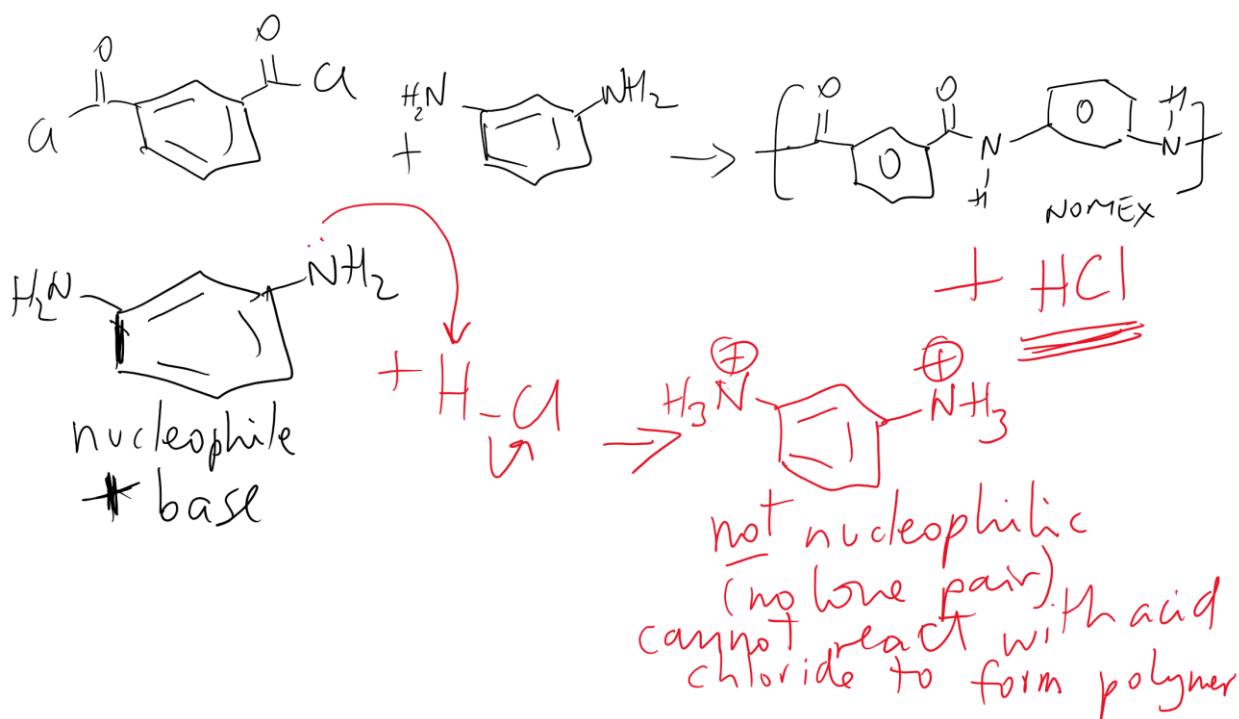
c) Draw the mechanism of the reaction that takes place in b).

Assumes TEA $\wedge$ N is used



- d) Normally a tertiary amine would be added to the mixture (like triethylamine, TEA). Why? What is the impact on MW if TEA is not added?

For full marks: The formation of polyamides by reaction of acid chlorides and amines has HCl as a by-product. This is problematic because the amine is also a good base, which will react with HCl as shown below. The ammonium ion product of this acid-base reaction is not a nucleophile and cannot take part in the polymerisation. The overall effect is to reduce the amount of amine, which will have an impact on the stoichiometric balance, reducing the DP (as per general Carothers equation). Therefore, a tertiary amine, such as TEA is used. The tertiary amine cannot form an amide, because it doesn't have an H attached to N to lose, but it's still basic to react with the HCl.



- e) Explain the differences of properties between Kevlar® and Nomex® based on the structure of the polymer

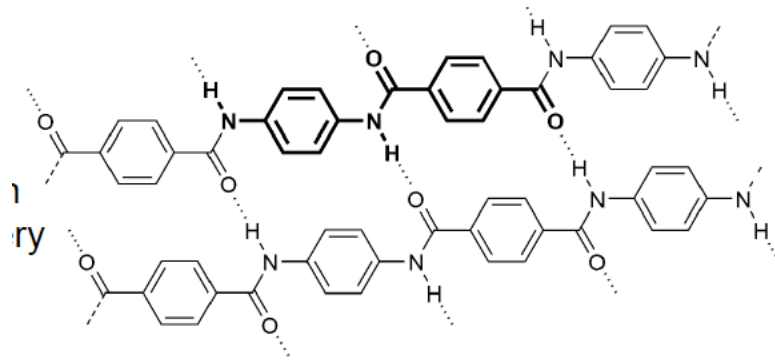
Some examples of thermal properties:

Kevlar Tg: 307-347 °C Decomposition temperature 427-482 °C

Nomex: Tg: 240-280 °C Decomposition temperature 400 °C

The para linkages in Kevlar, makes the structure of the chain can achieve a higher extension than in Nomex. Therefore, the material has a higher degree of crystallinity, lower free-volume, and the resulting fibres are extremely strong.

Kevlar:



Nomex: In Nomex, the meta linkages create a “kink” in the main chains, which introduces more free-volume between the chains. This reduces the packing, the degree of crystallinity that can be achieved with Nomex fibres. Therefore, the thermal and mechanical properties of Nomex are not as good as in Kevlar. However, the material remains extremely heat and chemical resistant.

